SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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QUESTIONS

- **Q1** Web Browser Back Navigation — 10 marks ✓
- **Q2** University Course Enrollment Manager — 10 marks ✓
- **Q3** Railway Parcel Dispatch Tracking System — 10 marks ✓

3/3 answered

Q1 Web Browser Back Navigation

10 marks

A web browser stores recently visited pages so that users can navigate backward.

Requirements:

Add visited page URLs using push()
Perform the Back operation using pop()
Display the current page using peek()

Your Code

```java
import java.util.Stack;
import java.util.Scanner;
public class BrowserBackNavigation {
    public static void main(String[] args) {
        Stack<String> history = new Stack<>();
        Scanner sc = new Scanner(System.in);
        history.push("google.com");
        history.push("wikipedia.org");
        history.push("simatslab.saveetha.com");
        history.push("github.com");
        System.out.println("Current Page: " + history.peek());
        System.out.println("\nGoing Back");
        history.pop();
        System.out.println("Current Page: " + history.peek());
        System.out.println("\nGoing Back");
        history.pop();
        System.out.println("Current Page: " + history.peek());
        System.out.println("\nRemaining Browser History:");
        for (String url : history) {
            System.out.println(url);
        }
        sc.close();
    }
}
```

Run Save

Input

stdin input

Output

```
Current Page: github.com

Going Back
Current Page:
simatslab.saveetha.com
```

QUESTIONS

3/3 answered

Q2 University Course Enrollment Manager

10 marks

A university maintains a list of students enrolled in a course. Students who withdraw must be removed from the list.

Requirements:

Add student names to an ArrayList

Traverse the list using an Iterator

Remove withdrawn students and display the updated enrollment list

Your Code

```java
import java.util.ArrayList;
import java.util.Iterator;
public class UniversityCourseEnrollmentManager {
    public static void main(String[] args) {
        ArrayList<String> enrolledStudents = new ArrayList<>();
        enrolledStudents.add("Aarav");
        enrolledStudents.add("Priya");
        enrolledStudents.add("Rohan");
        enrolledStudents.add("Sneha");
        enrolledStudents.add("Kabir");
        System.out.println("Initial Enrollment List:");
        for (String student : enrolledStudents) {
            System.out.println(student);
        }
        ArrayList<String> withdrawnStudents = new ArrayList<>();
        withdrawnStudents.add("Rohan");
        withdrawnStudents.add("Sneha");
        Iterator<String> iterator = enrolledStudents.iterator();
        while (iterator.hasNext()) {
            String currentStudent = iterator.next();
            if (withdrawnStudents.contains(currentStudent)) {
                iterator.remove();
                System.out.println("\nRemoved (Withdrawn): " + currentS
            }
        }
        System.out.println("\nUpdated Enrollment List:");
        for (String student : enrolledStudents) {
```

Input

stdin input

Output

```
Initial Enrollment List:
Aarav
Priya
Rohan
Sneha
Kabir
```

QUESTIONS

3/3 answered

Q3 Railway Parcel Dispatch Tracking System

10 marks

A railway parcel office loads parcels onto a dispatch container. The last loaded parcel must be unloaded first, and parcel details must be searchable.

Requirements:

Store parcel IDs in a Stack

Store parcel ID and destination in a HashMap

Unload parcels using pop() and display the destination of the unloaded parcel

Your Code

```java
import java.util.*;
public class RailwayParcelDispatch {
    public static void main(String[] args) {
        Stack<Integer> parcelStack = new Stack<>();
        HashMap<Integer, String> parcelMap = new HashMap<>();
        parcelStack.push(101);
        parcelMap.put(101, "Chennai");
        parcelStack.push(102);
        parcelMap.put(102, "Bangalore");
        parcelStack.push(103);
        parcelMap.put(103, "Hyderabad");
        parcelStack.push(104);
        parcelMap.put(104, "Mumbai");
        System.out.println("Parcels loaded onto the dispatch container:
        System.out.println(parcelStack);
        System.out.println("\nUnloading parcels (Last In, First Out):")
        while (!parcelStack.isEmpty()) {
            int unloadedParcelId = parcelStack.pop();
            String destination = parcelMap.get(unloadedParcelId);
            System.out.println("Unloaded Parcel ID: " + unloadedParcel
        }
        System.out.println("\nAll parcels have been unloaded and dispat
    }
}
```

Input

```
stdin input
```

Output

```
Parcels loaded onto the dispatch
container:
[101, 102, 103, 104]

Unloading parcels (Last In,
First Out):
```